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CALCIUM PHOSPHATE IN PHARMACEUTICAL PRODUCT DEVELOPMENT

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INTRODUCTION

Calcium phosphate is a widely used pharmaceutical excipient as binder and filler in solid oral dosage forms which include compressed tablets and hard gelatin capsules. A review of the marketed tablets and capsules indicated that several of them contain calcium phosphate. Some of the commonly used dosage forms are Benadryl®, Sudafed®, Decadron®, Norvasc®, Ex-Lax®, and Gas-X®. The wide acceptance of calcium phosphate is primarily due to its excellent flowability and compressibility characteristics. The relatively low cost and its elemental calcium value adds to its popularity. It is mainly used as anhydrous and dihydrate forms of dibasic calcium phosphate, and tribasic calcium phosphate. While the calcium phosphate dihydrate is commercially available from different sources, Emcompress® from Mendell and Di-Tab® from Stauffer are very popular. The present chapter focuses on the physical and chemical properties of these calcium phosphates, flow property measurements, compressional behavior and the forces

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involved, tablet compression studies, and certain excipient incompatibilities. Calcium phosphates have also been used as anti-caking or flocculating agents in suspensions, and adsorbent or thickening agents in ointments and creams. However, the present chapter is focused towards its use in solid dosage forms. Since the present book is intended for a multidisciplinary readership, an introduction is provided for the method of preparation of tablets, type of tablets, function of tablet excipients, and the evaluation of compressed tablets.

METHODS OF TABLET PREPARATION

Tablets are oral solid unit dosage forms in which one usual dose of the drug has been accurately placed. When compared to liquid oral dosage forms, tablets are less expensive to ship, easy to store, least likely to undergo microbial attack, and offer better shelf-life stability. Further, tablets offer greatest dose precision and the least content variability. Tablet technology lends itself to the design of a variety of therapeutic products with a broad range of drug release characteristics, i.e., immediate, controlled-release, delayed-release, release-in stages, etc.

In addition to the active or therapeutic ingredient(s), tablets contain a variety of other inert ingredients (or additives) with specialized functions. These ingredients contribute to the physical form, stability, taste, appearance and overall effectiveness of the tablet dosage form. Therefore, the excipients may be included in tablet formulations to improve physical appearance, enhance stability and facilitate disintegration process following administration. The most common tablet excipients include diluents (or fillers), binders, anti-adherents, glidants, lubricants, colorants, and disintegrating agents. Although these ingredients, in general, are non-therapeutic, some of them have shown to influence the release of the drug substances which in turn affects the clinical outcome. The production methods employed may also affect the physiological availability of the drug from tablets. In light of these reasons, the personnel involved in large scale production of tablets should not only consider the physiological availability of tablet formulation but also the non-therapeutic excipients, processes, and equipment. Therefore care must be exerted in the selection as well as evaluation of excipients to ensure that the therapeutic efficacy of the active ingredient(s) will not be affected.

Compressed tablets are prepared by three methods including: A. dry granulation, B. wet granulation, and C. direct compression. Of the three methods, wet granulation is the most widely used method for the production of tablets.

Wet Granulation

Substances having crystalline structure may be directly compressed into a tablet without additional treatment. However, most substances require pretreatment to impart free flowability in the tablet machine and subsequent compression into tablets. The pretreatment in which fine powders are converted to free flowing